LearnGeeta: A Flutter-Based Gamified Mobile Platform for Bhagavad Gita Learning

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***Abstract*—This paper presents LearnGeeta, a Flutter mobile application for Bhagavad Gita learning that combines reading, audio support, and gamified practice. The implementation uses Supabase for authentication and progress persistence, local JSON assets for deterministic scripture delivery, and twelve game modes for recall, sequencing, ethical reasoning, and conceptual classification. We analyze the shipped system architecture, data flow, and reward logic directly from the codebase, then connect each subsystem to evidence from a curated set of fifteen downloaded research papers. The paper also formalizes core in-app algorithms, including level progression, streak updates, timer-driven rounds, and karma-based branching outcomes. The resulting analysis shows how LearnGeeta balances practical mobile constraints with pedagogy for Sanskrit and philosophy learning, while also identifying current limits such as sparse integration testing, no explicit spaced repetition scheduler, and tension between extrinsic rewards and intrinsic scripture study.**

***Index Terms*—Flutter, mobile learning, gamification, Sanskrit learning, Bhagavad Gita, serious games, text to speech.**

I. Introduction

Mobile learning systems can support daily study habits, but scripture learning adds specific requirements: reliable text integrity, language scaffolding, and motivation that lasts beyond novelty. LearnGeeta addresses this with a single app that combines reading and guided play. The app ships with chapter and verse content, transliteration, translation, text to speech playback, and a progression loop built around experience points (XP), levels, streaks, and achievements.

The design direction matches current gamification and mobile learning findings in the downloaded literature pack [1], [5], [12], [14]. The app also uses scenario-based and decision-based game modes, aligning with branching and serious game evidence [2], [7], [8], [13]. For language and accessibility, the product applies multimodal presentation (text plus audio) supported by research on text to speech and Sanskrit digital learning contexts [3], [4], [9]. At the same time, the evidence is not one-sided: print-vs-digital retention tradeoffs remain important [6], and cognitive load needs active control in game design [11].

This paper contributes three things:

- A code-grounded architecture report for the current LearnGeeta implementation.
- A mathematical formulation of key progression and gameplay algorithms used in production code.
- A research-to-feature synthesis using all fifteen downloaded references [1]-[15].

II. Reference corpus and methodology

The analysis uses the local LearnGeeta codebase, the included IEEE conference template, and a downloaded set of fifteen papers stored under the project reference pack. Selection was feature-driven: gamification impact [1], [5], [12], [14], branching pedagogy [2], serious game design [7], [8], [13], multimodal and text-to-speech support [3], Sanskrit learning context [4], [9], [10], cognitive load guidance [11], digital media retention tradeoffs [6], and adaptive platform outcomes [15].

Instead of reporting a new user trial, this paper performs a system-level implementation study. Evidence is linked to concrete shipped features: authentication, reader flow, gameplay loops, progress persistence, and tests.

III. System architecture of LearnGeeta

A. Technology stack

LearnGeeta is built in Flutter with a modular project structure. Core runtime dependencies include *supabase_flutter* for auth and cloud persistence, *shared_preferences* for local state, *flutter_tts* for audio playback, and *flutter_dotenv* for environment loading. The application entry point initializes Supabase and routes users through an authentication gate before loading the tabbed learning experience.

B. Layered design

The implementation follows a practical layered approach:

- Presentation layer: screens for Home, Learn, Play, and Progress.
- Controller/state layer: ChangeNotifier-based controllers for screen logic and sync events.
- Repository layer: auth, profile, progress, and content repositories.
- Data layer: local Bhagavad Gita JSON corpus plus Supabase tables for progress and game stats.

A static dependency container wires repositories and services. XP award operations trigger progress sync notifications so level-up events propagate to UI without tight coupling.

C. Content and learning pipeline

Scripture data is loaded from a local JSON file containing 18 chapters and verse-level fields (text, transliteration, word meanings, and translation arrays). The repository includes fallback handling in case JSON loading fails. This local corpus avoids runtime content drift and keeps scripture output deterministic, a practical response to concerns raised in Sanskrit AI tooling discussions [10].

The Learn module persists bookmarks (last chapter and verse) in local storage. This supports short recurring study sessions rather than requiring long uninterrupted reading blocks.

D. Navigation and game composition

Bottom navigation uses an IndexedStack for stable tab state. The Play module registers twelve games:

- Shloka Match
- Verse Order
- True or False
- Dharma Choices
- Missing Word Mantra
- Chapter Quest
- Wheel of Gunas
- Krishna Memory Cards
- Battlefield Debate
- Krishna Says
- Shloka Speed Run
- Karma Path

This blend includes recall drills, sequence reconstruction, speed rounds, conceptual classification, and branching moral scenarios. That mix aligns with evidence that engagement gains are stronger when systems combine light gamification with deeper serious-game mechanics [7], [8], [13].

IV. Mathematical and Algorithmic Formulation

This section formalizes the progression logic that is actually implemented.

A. Level function and XP accumulation

The global level is computed from total XP using a linear policy:

$$L(x) = \left\lfloor \frac{x}{100} \right\rfloor + 1, \tag{1}$$

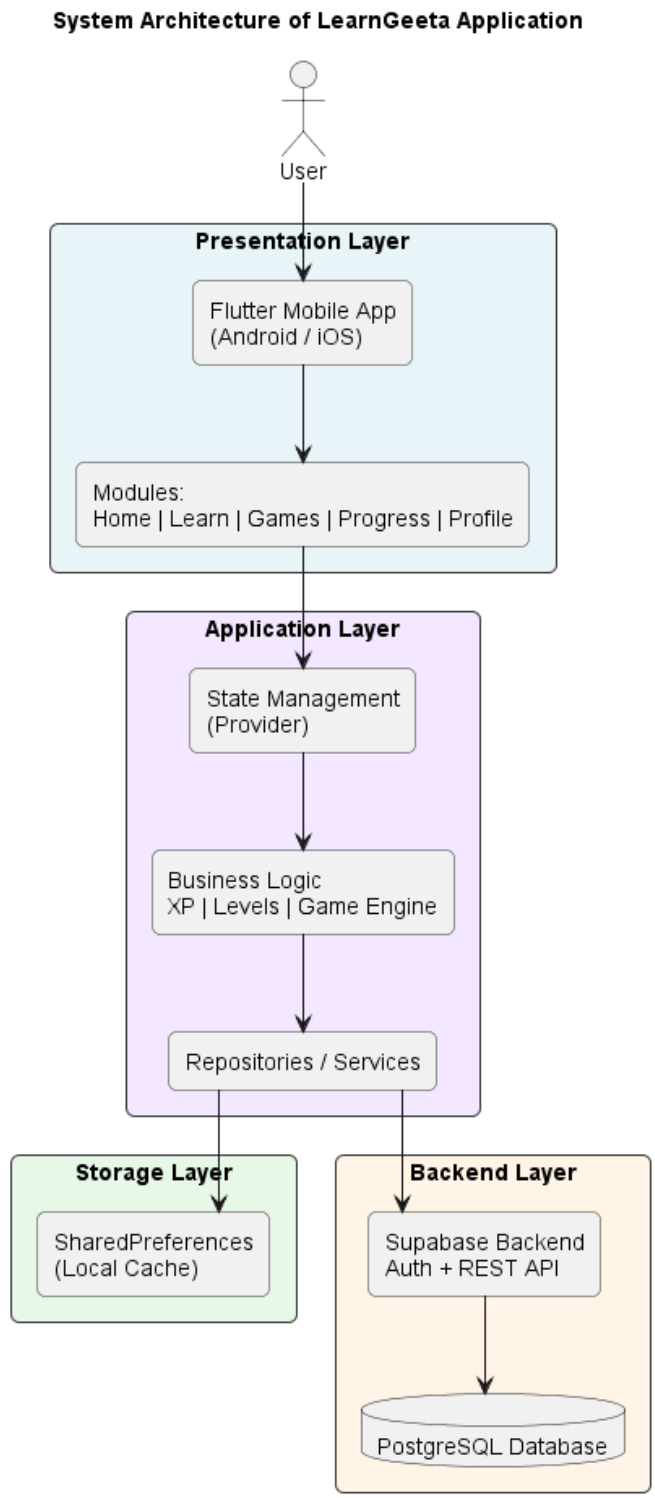

Fig. 1. System Architecture of LearnGeeta Application

where x is total accumulated XP.

Each game awards XP with rule-specific bonuses. For example:

$$xp_{tf} = 10 + 5\,\mathbf{1}_{hard} + 5\,\mathbf{1}_{s\geq 3}, \tag{2}$$

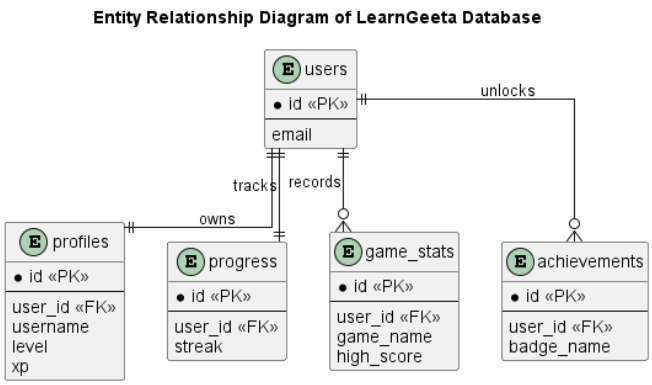

Fig. 2. Entity Relationship Diagram of LearnGeeta Database

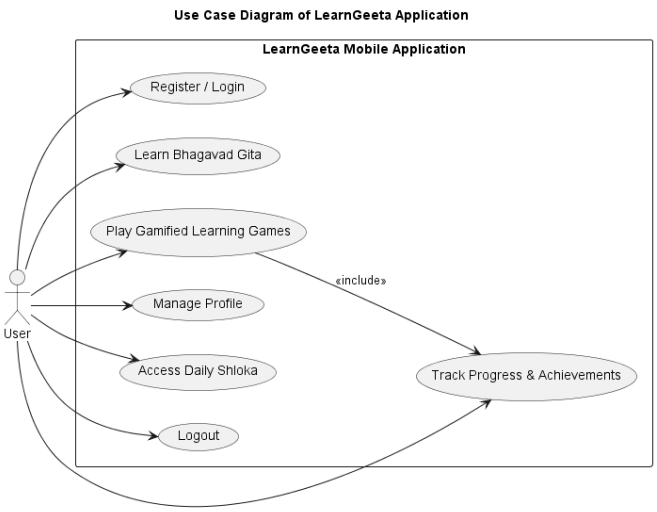

Fig. 3. Use Case Diagram of LearnGeeta Application

for True/False, and

$$xp_{sr} = 8 \cdot \min(\max(m, 1), 3), \tag{3}$$

for Shloka Speed Run, where m is combo multiplier.

In Karma Path, per-choice XP and ending XP are piecewise by karma score, with higher rewards for consistently sattvic choices.

B. Daily streak update rule

Let s_t be streak on day t, and Δd be whole-day distance from the last active date:

$$s_t = \begin{cases} 1, & \text{first activity}, \\ s_{t-1} + 1, & \Delta d = 1, \\ 1, & \Delta d > 1, \\ s_{t-1}, & \Delta d = 0. \end{cases} \tag{4}$$

This logic preserves same-day activity and resets only after missed days. The streak mechanism maps to habit-formation evidence [15] while still requiring UX safeguards against streak-loss frustration [6], [11].

1: Fetch current progress $(xp, level, streak, last_active)$
2: Add earned XP: $xp \leftarrow xp + \Delta xp$
3: Recompute level using (**??**)
4: Compute day difference Δd
5: **if** $\Delta d = 1$ **then**
6: $\quad streak \leftarrow streak + 1$
7: **else if** $\Delta d > 1$ **then**
8: $\quad streak \leftarrow 1$
9: **end if**
10: Persist updated progress to Supabase
11: Notify UI listeners for progress and level-up events

Fig. 4. Progress update pipeline as implemented by repository plus XP service layers.

TABLE I
EXAMPLES OF GAME LOOP PARAMETERS IN CURRENT CODE

Mode	Implemented rule
Shloka Match	30 s timer, difficulty by level bands (1-3 easy, 4-7 medium, 8+ hard), streak bonus tiers.
True/False	15 s base timer with level-dependent reduction, non-repeating question cycle before reset.
Shloka Speed Run	45 s round, question subset filtered by level, combo multiplier cycles across bounded range.
Karma Path	Randomized start node, signed karma accumulation, ending class selected by karma interval.

C. Deterministic daily shloka indexing

Home screen uses a date-based index into a fixed shloka pool:

$$i_d = \left\lfloor \frac{t_d}{86{,}400{,}000} \right\rfloor \bmod N, \tag{5}$$

where t_d is day timestamp in milliseconds and N is pool size. This gives stable day-level rotation without server calls.

D. Mode-specific loop tuning

These loops match broad findings that immediate feedback, challenge scaling, and repeated retrieval improve engagement and retention [1], [5], [12], [14].

V. FEATURE-TO-EVIDENCE MAPPING

Table **??** summarizes how the app modules align with the downloaded references.

Together, these sources support the core design while also exposing risk areas: reward overemphasis, shallow streak behavior, and insufficient personalization if spacing and adaptation are not made explicit [6], [11], [15].

TABLE II
MAPPING OF LEARNGEETA MODULES TO DOWNLOADED RESEARCH EVIDENCE

App subsystem	Why it matters in LearnGeeta	Primary downloaded references
XP, levels, achievements, streaks	Sustains repeat usage and visible progression during long scripture learning timelines.	[1], [5], [12], [14], [15]
Branching and ethics games	Supports decision reasoning and consequence-based reflection, not only fact recall.	[2], [7], [8], [13]
Reader + transliteration + TTS	Reduces decoding friction for Sanskrit beginners and broadens accessibility.	[3], [4], [9]
Bhagavad Gita value framing	Connects technical design to emotional and cognitive development goals.	[10]
Cognitive load and pacing choices	Justifies timer, difficulty, and UI simplification tradeoffs in game rounds.	[11]
Digital-first delivery caveat	Warns that deep retention can differ from print workflows without active recall support.	[6]

VI. IMPLEMENTATION VERIFICATION AND QUALITY STATUS

A. What is covered now

The repository includes unit and widget tests for important user-facing behavior:

- Auth controller checks password strength before backend calls and verifies successful sign-in behavior.
- Play screen test verifies that selecting a game card opens the expected screen.
- Bottom navigation test confirms IndexedStack state persistence across tab changes.
- Input widget test validates password field visibility toggle behavior.

This gives useful guardrails for auth and navigation regressions, which are common failure points in mobile apps.

B. Current technical limits

Three implementation gaps are visible from code inspection:

- No explicit spaced repetition scheduler that reintroduces verses by memory decay probability.
- Limited integration testing for Supabase sync, offline retries, and cross-device conflict behavior.
- Game content is largely static in local model files, which limits continuous curriculum updates.

These are practical next steps rather than architectural failures. The existing structure can support them with moderate refactoring.

VII. DISCUSSION

LearnGeeta shows a realistic pattern that many educational apps struggle to reach: it is not only a reader and not only a game hub. The shipped system links text study, audio support, and repeated retrieval practice in one flow. That is a strong base.

The main design tension is philosophical and behavioral. Bhagavad Gita learning often emphasizes inner discipline, while mobile loops often rely on visible rewards. The app currently leans on XP and streaks to maintain consistency, which is effective for habit formation [1], [12], [14], [15], but should be balanced with lower-pressure study paths to avoid metric chasing [6], [11]. The reference set also suggests a direction for next iterations: explicit adaptive sequencing, richer scenario depth, and better handling of difficult verses instead of equal random exposure [2], [7], [8], [13].

VIII. CONCLUSION

This paper documented the LearnGeeta application as implemented, not as a hypothetical design. The app already integrates a meaningful combination of scripture content delivery, language support, gamified reinforcement, and cloud-backed progress tracking. Formal analysis of the code-level rules shows clear, deterministic progression logic and practical gameplay tuning.

The downloaded research corpus supports most design decisions and also points to the next technical milestone: turning todayś gamified practice into a more adaptive learning engine with robust offline-first testing and memory-aware scheduling. With those additions, LearnGeeta can move from a strong educational app to a high-confidence digital learning platform for sustained Bhagavad Gita study.

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